



**National Laboratory
of the Rockies**



U.S. DEPARTMENT
of **ENERGY**

Office of Electricity

Multidimensional uncertainty assessment in a capacity expansion model for the U.S. power system

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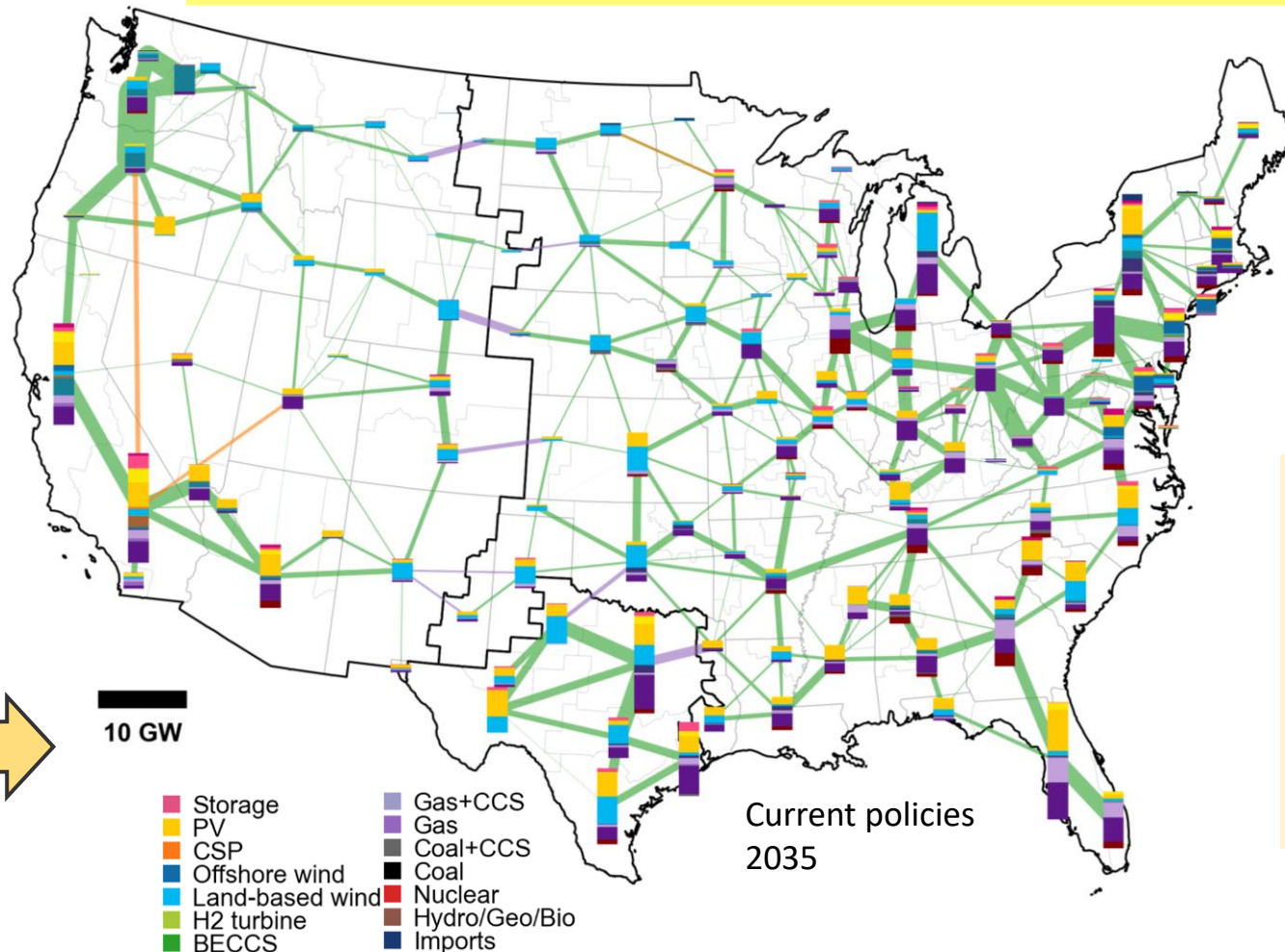
Capacity expansion + resource adequacy modeling

2

Inputs

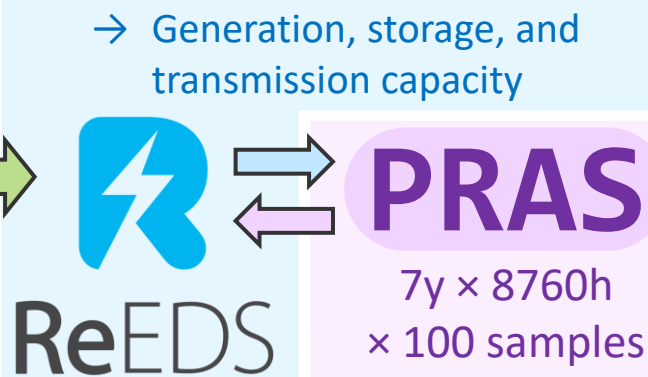
- Demand (hourly) projections
- Cost & fuel price projections
- Renewable energy **availability**
- State & federal **policies**
- **Technology** availability
- Existing & **planned capacity**

What mix of **generation and transmission** capacity would meet **electricity demand** and **policy + operational constraints** at **least cost** through 2050?



Outputs

- Co-optimized regional **capacity and operations** of:
 - Generation
 - Transmission
 - Storage



Capacity expansion

Resource adequacy

- ← Is the system resource adequate?
- ← What conditions should be modeled to ensure reliability?

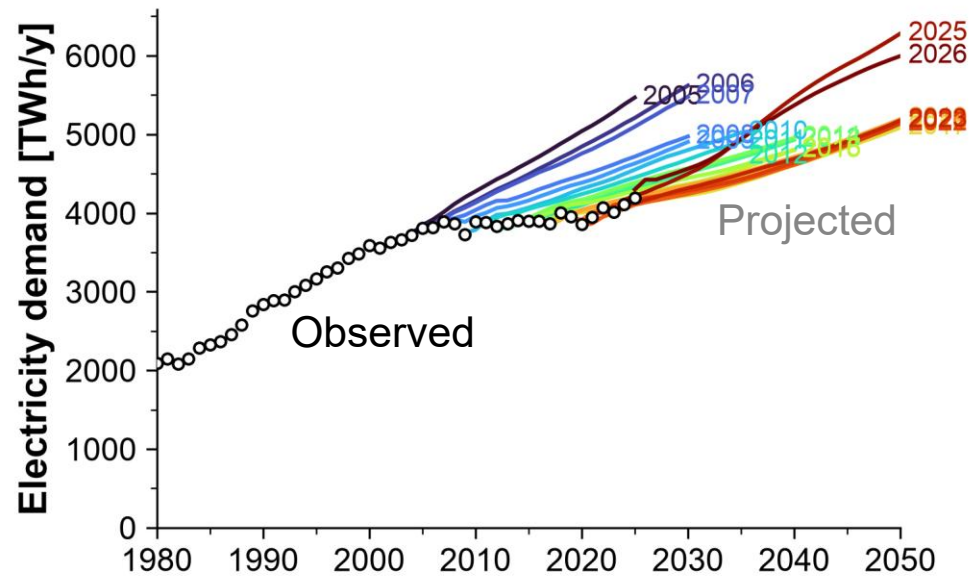
<https://github.com/ReEDS-Model/ReEDS>

<https://github.com/NatLabRockies/PRAS>

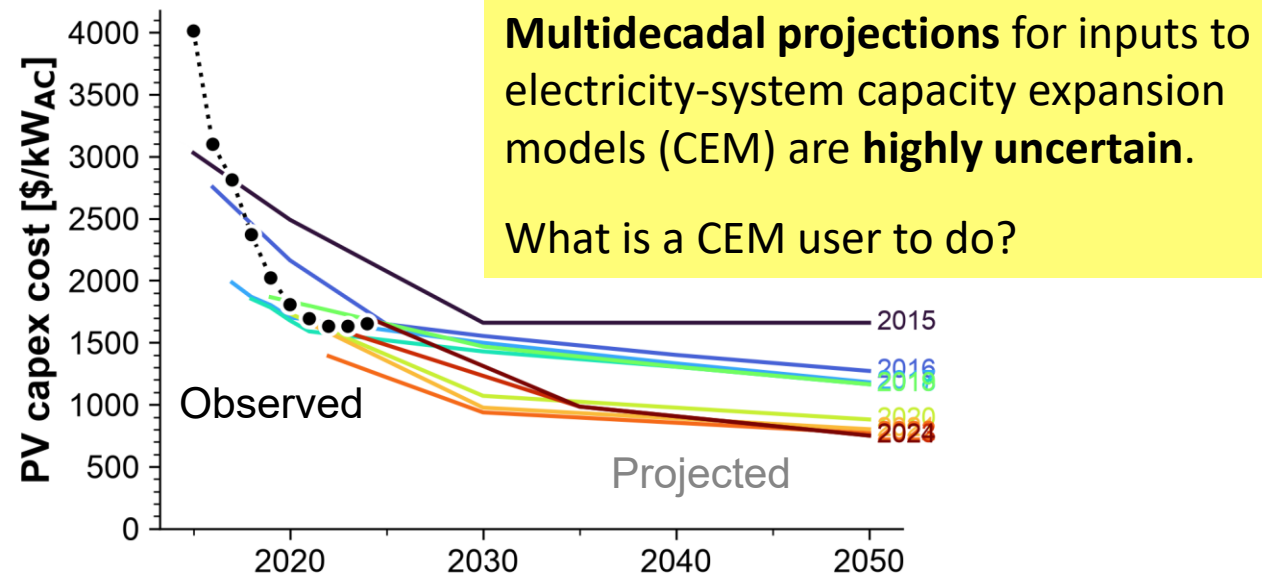
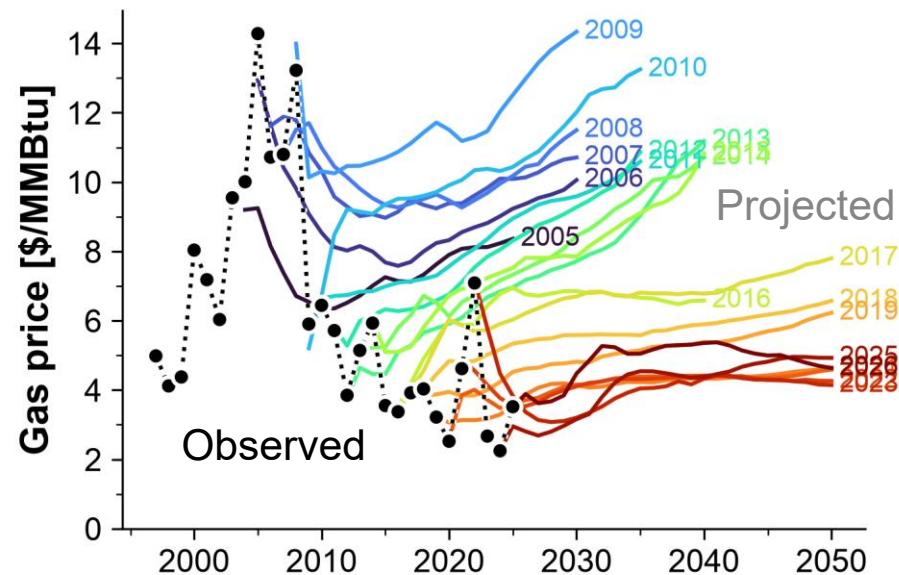
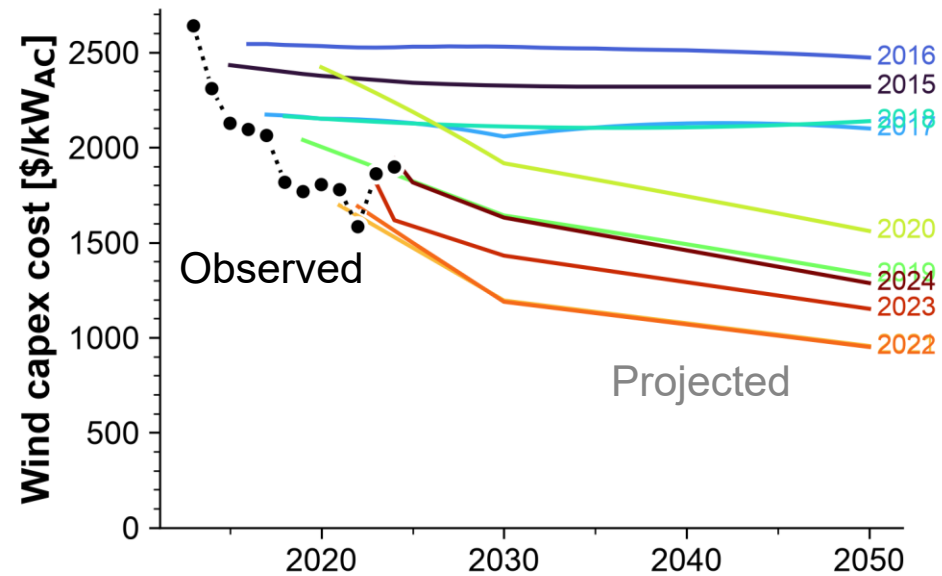
Predictions are hard

3

EIA AEO 2005–2026, central case projections



ATB 2015–2024, central case projections

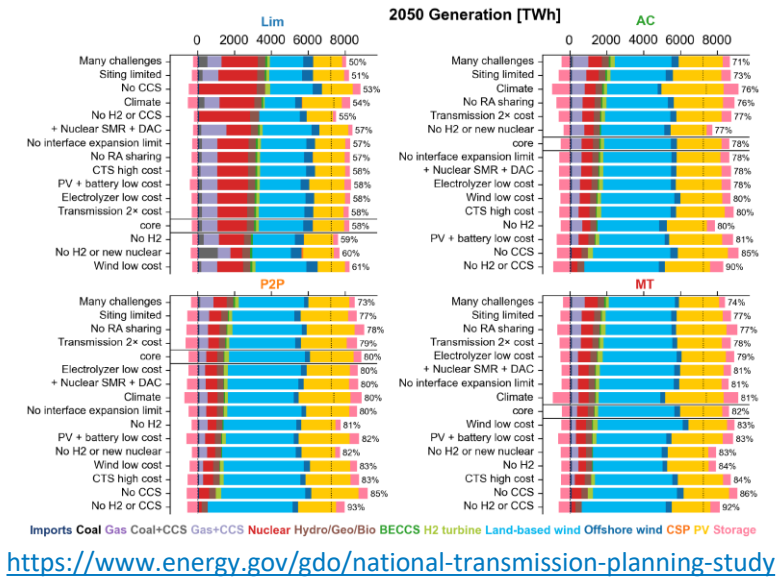


Multidecadal projections for inputs to electricity-system capacity expansion models (CEM) are highly uncertain.

What is a CEM user to do?

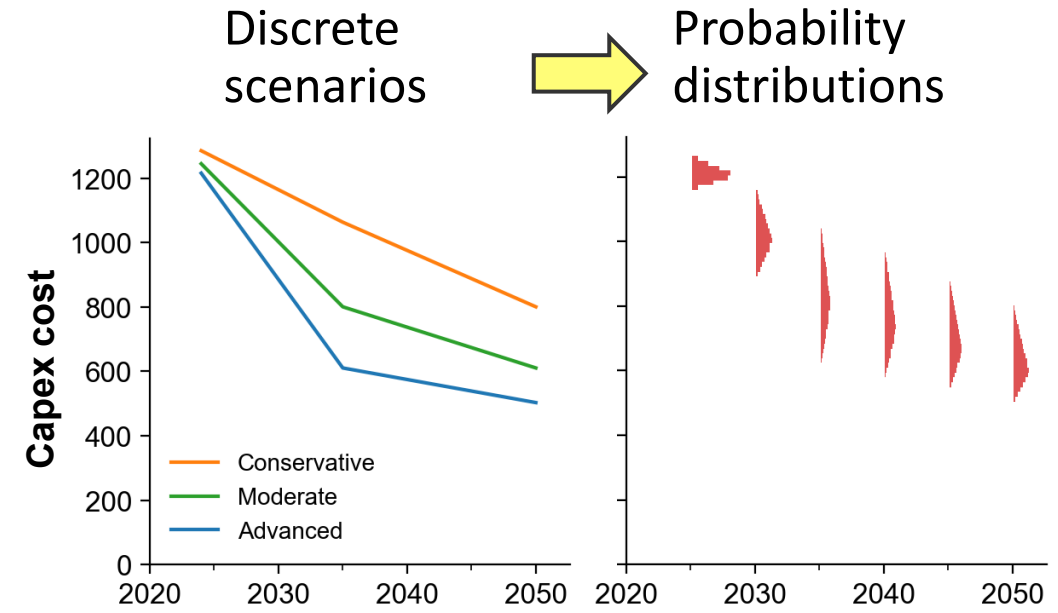
Three approaches to uncertainty quantification

Scenarios



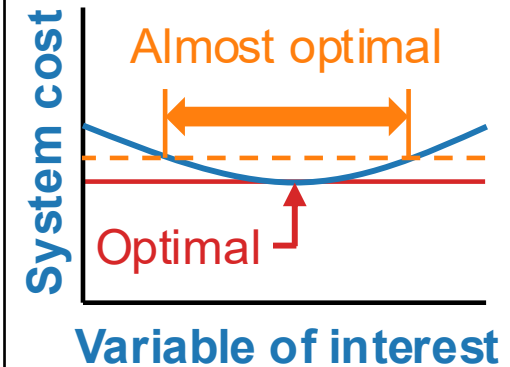
“What are some discrete alternative narratives?”

Monte Carlo Sampling



“How does uncertainty in input assumptions propagate to uncertainty in output results?”

Modeling to Generate Alternatives



“What are other near-optimal solutions?”

This presentation:

1. Modeling framework: ReEDS + PRAS (capacity expansion + resource adequacy)
2. Monte Carlo sampling of input parameters
3. Which categories of input-parameter uncertainty matter most?

Monte Carlo sampling of input parameters

5

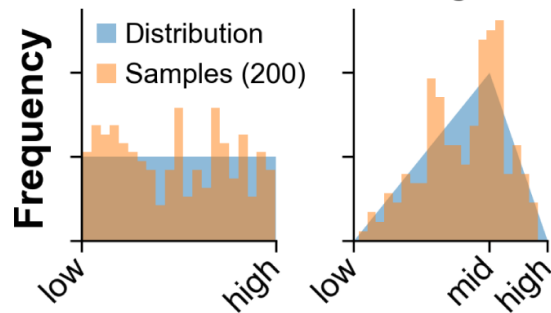
Monte Carlo sampling allows for multiple input dimensions to be treated probabilistically, with degrees of freedom related to:

1. Probability distribution **shape**
2. **Regional correlation** of samples

Shape of distribution

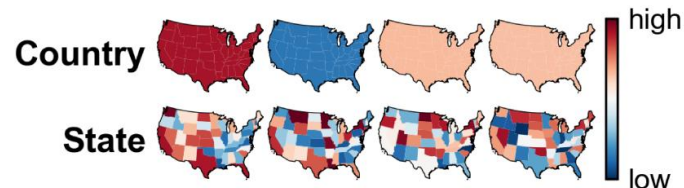
Flat

Triangular

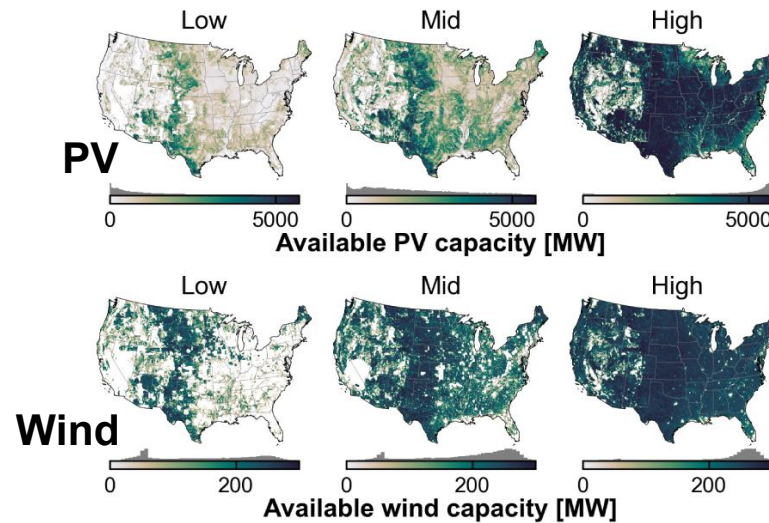


✕

Regional correlation across variables

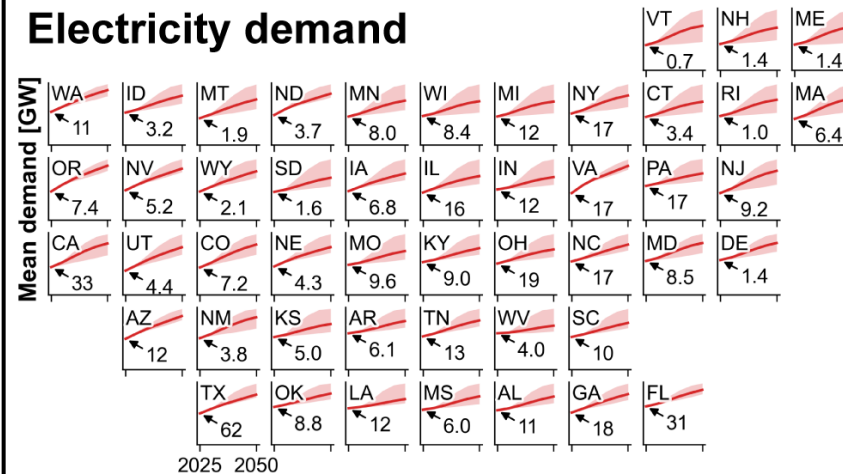


Siting availability



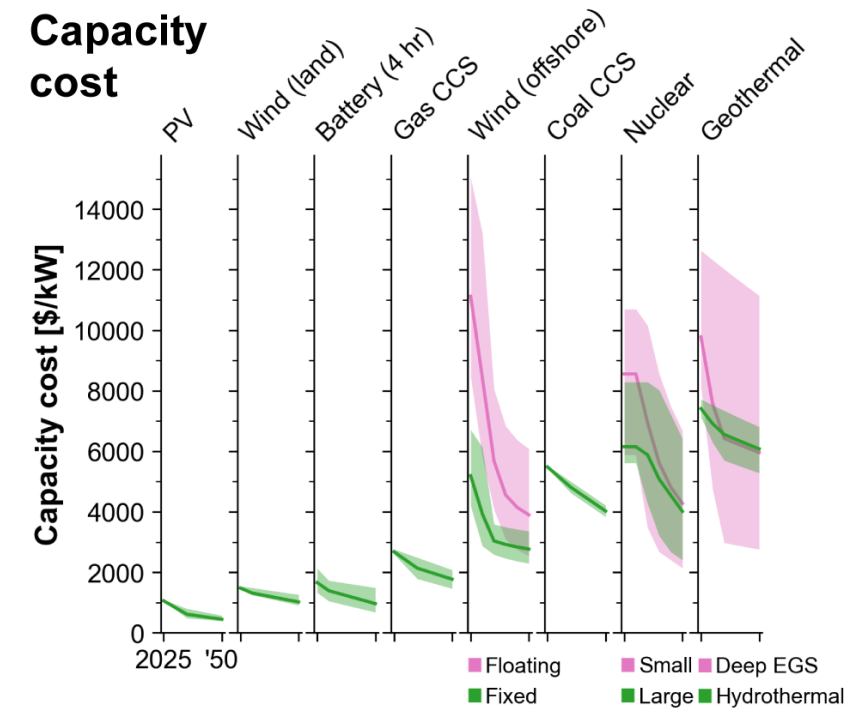
Lopez et al. 2025 <https://www.nlr.gov/docs/fy25osti/91900.pdf>

Electricity demand



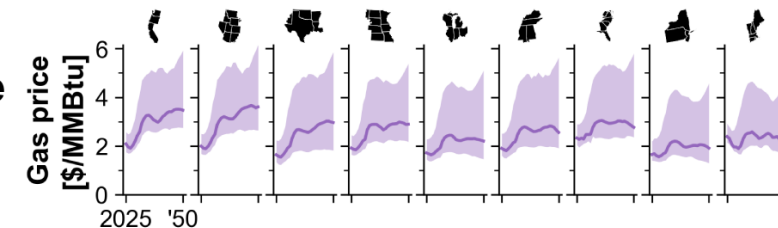
Jones et al. 2024 <https://www.evolved.energy/us-adp-2024>

Capacity cost



ATB 2024 <https://atb.nlr.gov/>

Gas price

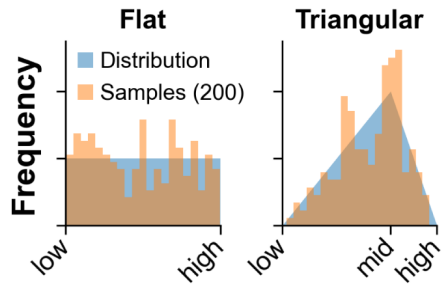


EIA AEO 2025 <https://www.eia.gov/outlooks/archive/aeo25/>

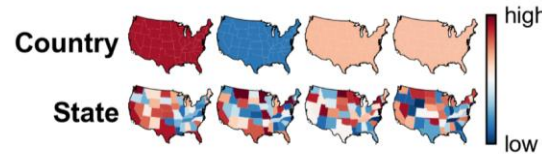
6 sets of ≥ 185 samples (1137 samples total)

6

Shape of distribution



Regional correlation across variables



Discrete **federal policy** alternatives relative to current policies:

1. Tax credit extension (+ TC)
2. No plant-level emission limits (– EL)

Name	Shape	Regionality	Policy	# runs
Tri country	triangular	country	current	194
Tri state	triangular	state	current	189
Flat country	flat	country	current	189
Flat state	flat	state	current	189
+ TC	flat	state	continued tax credits	185
– EL	flat	state	no emission limits	191

Address two research questions:

1. How do distribution shape, regional correlation, and policy uncertainty affect the **distribution of CEM results**?
2. How do **interactions** between input parameters affect systemwide metrics?

= 1137 full CEM+RA runs completed

1. These scenarios only capture some dimensions of **parametric** uncertainty, **not structural uncertainty**

1. How would a different balance of spatial & temporal **resolution** change the results?

2. What **objectives** besides cost minimization influence system evolution?

3. **ISO boundaries** have evolved over time; how might they change in the future?

2. Other **sources and magnitudes** of parametric uncertainty are not captured here

1. Demand & cost trajectories **outside the bounds** of our input data sources

2. **Interannual** weather variability beyond the 7 years used here for resource adequacy

3. Impacts of climate change on weather and demand profiles

3. A small selection of **alternative policies** are tested; many others are possible

4. Input cost projections are **less volatile** than historical observations

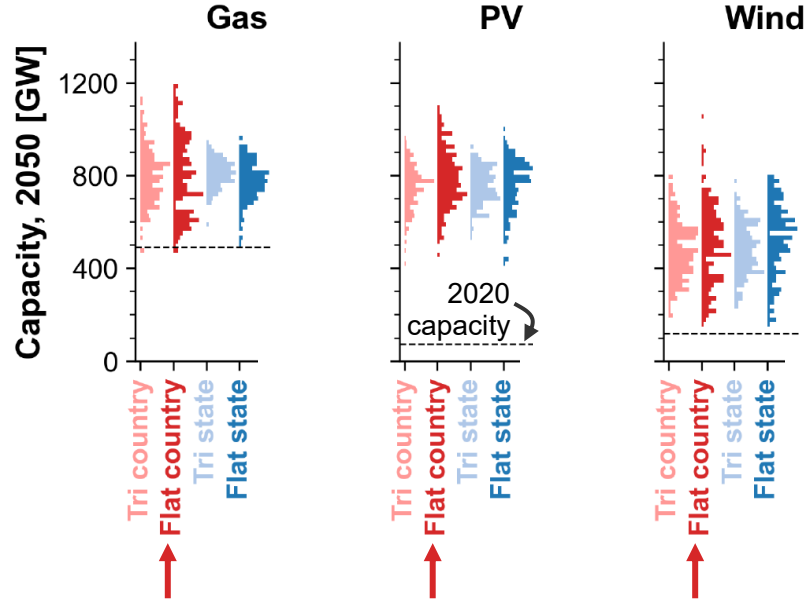
5. Our sources for low/mid/high projections **do not assign probabilities** to those projections

1. Use of **other input distributions** besides “flat” and “triangular” would change the output distributions

6. Our approach does **not** directly indicate a single recommended decision-making strategy

Impact of distribution, regionality, and policy

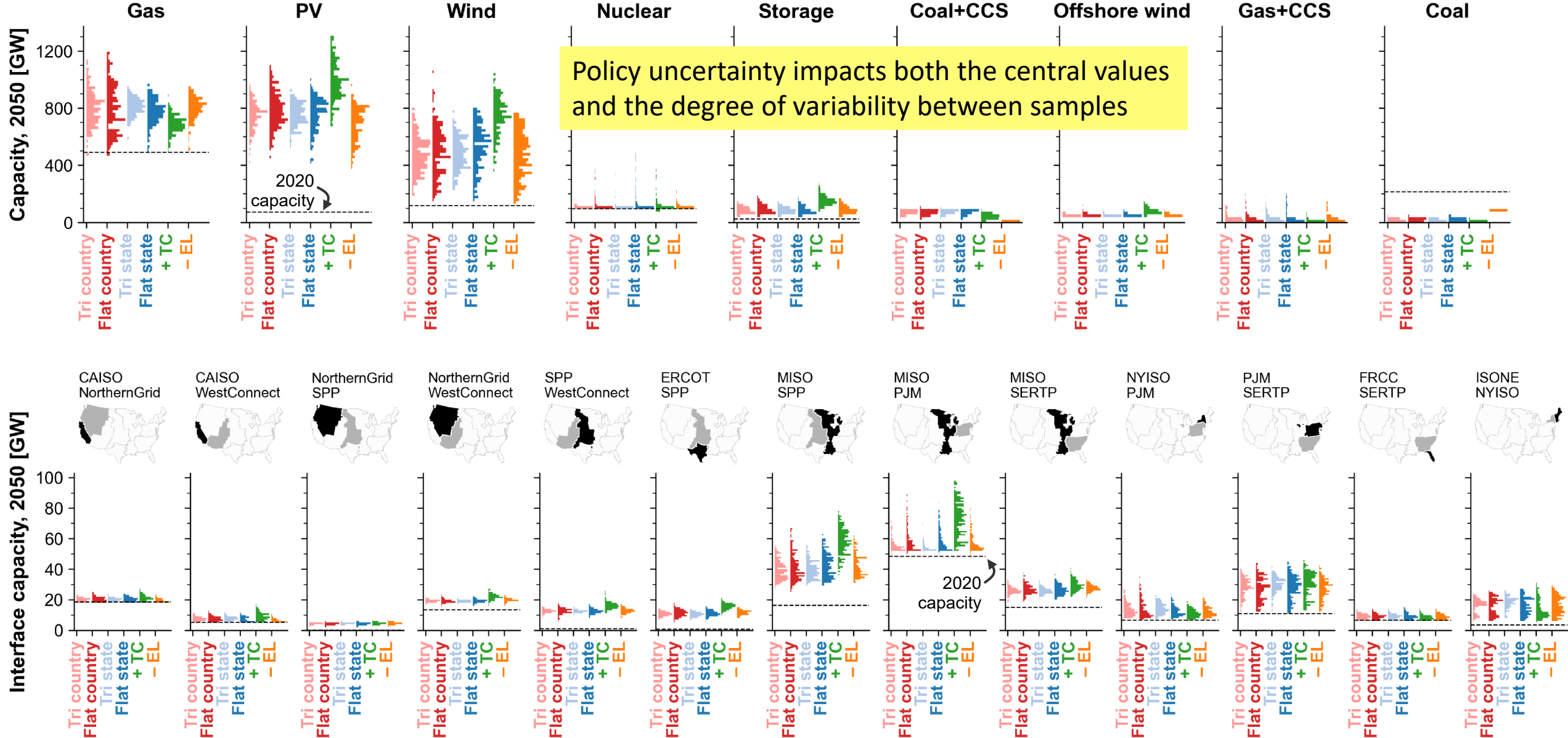
8

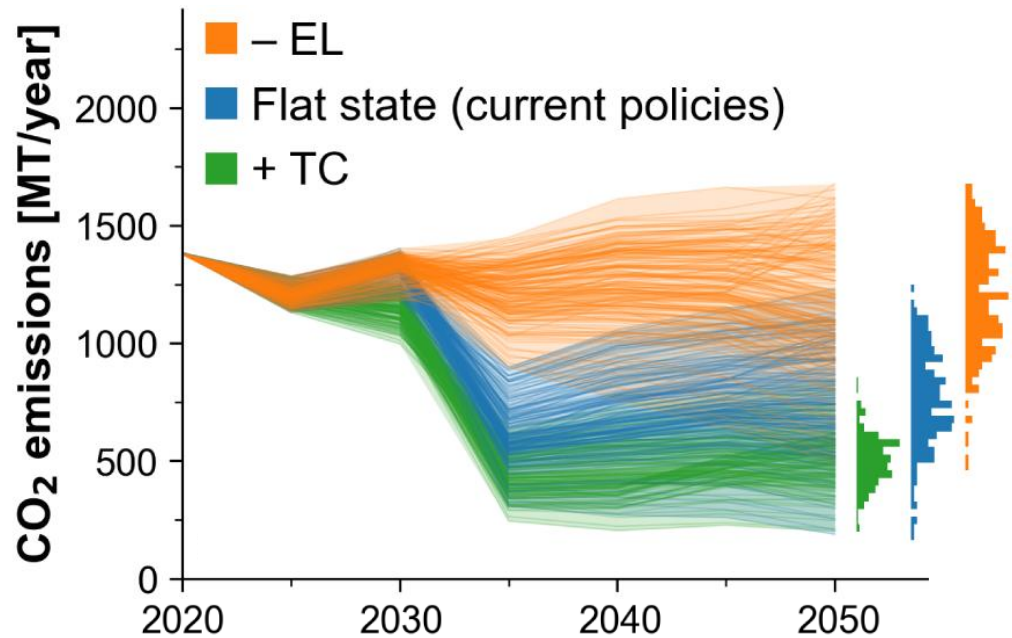


- Uniform distribution with full regional correlation (“**Flat country**”) produces the largest range in results
- Little change in central values across different distributions and degrees of regional correlation

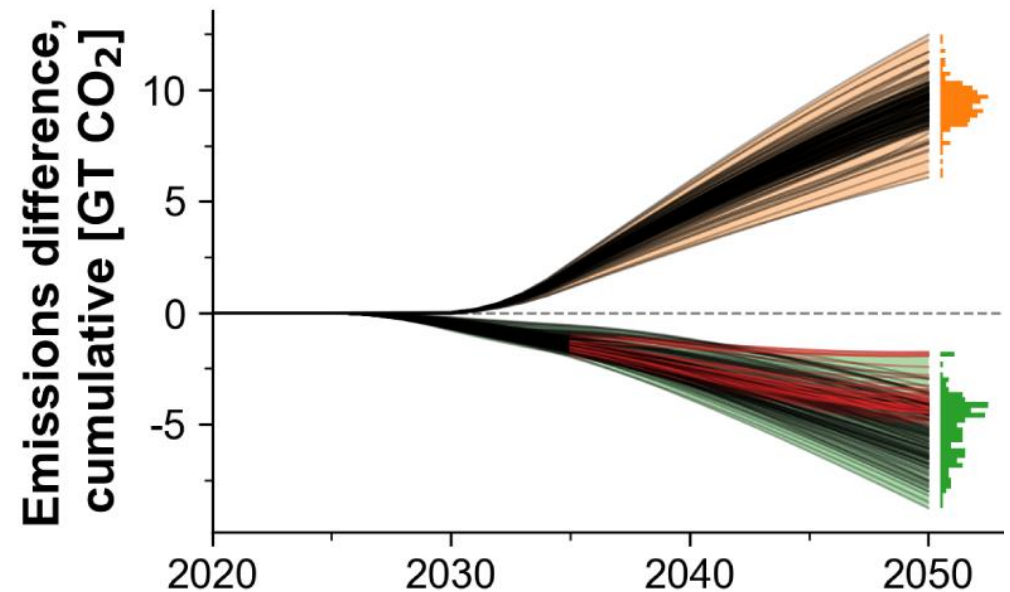
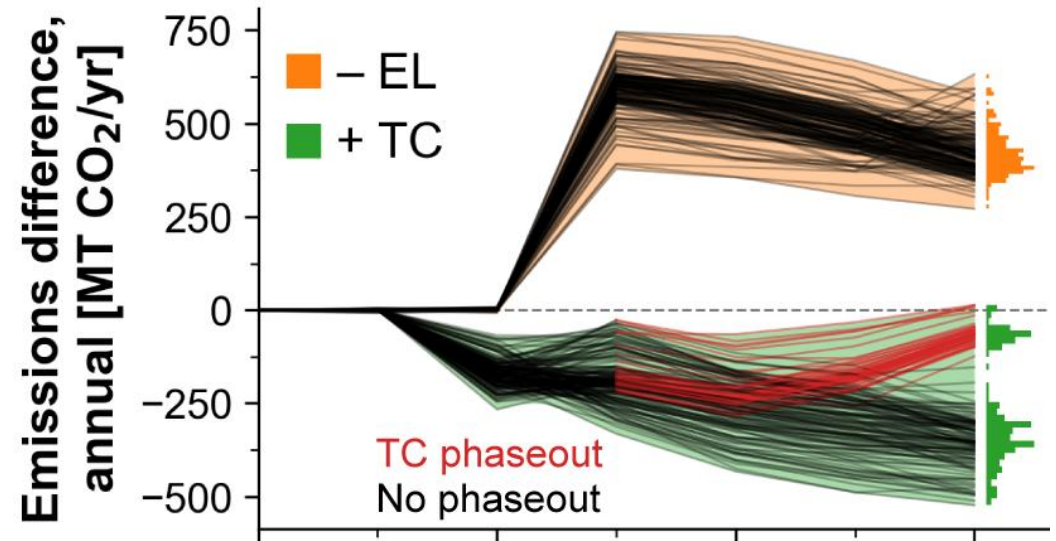
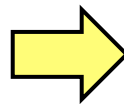
Impact of distribution, regionality, and policy

9

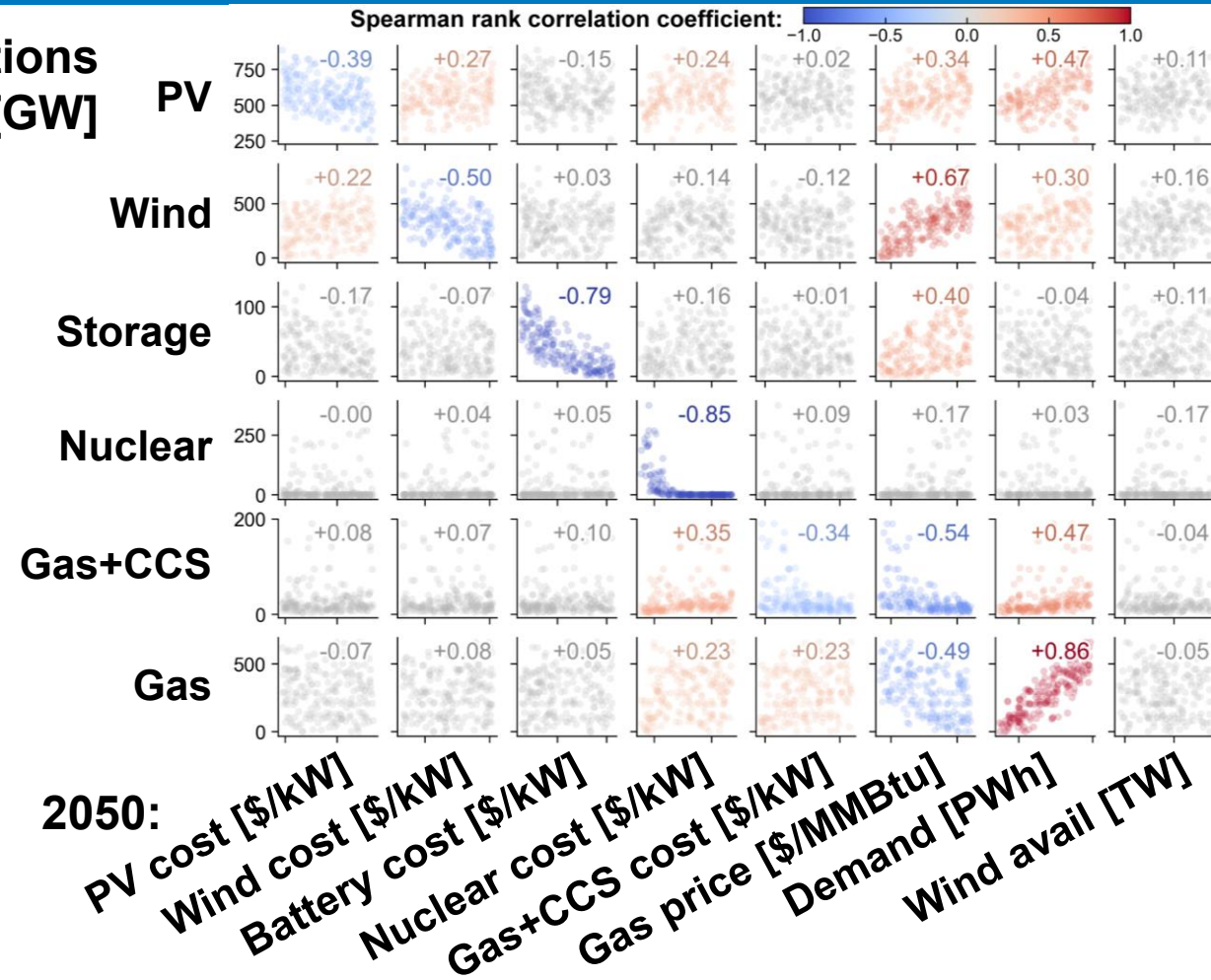




- Random number generator (RNG) seed is fixed between policy scenarios
- Can compare **paired Monte Carlo draws**, holding everything fixed except policy
 - Allows for **probabilistic assessment** of the impact of specific **policy interventions**



Capacity additions
through 2050 [GW]



- Flat distribution
- Country sampling (perfect regional correlation)
- 189 samples

Global sensitivity analysis
(varying all parameters together)
uncovers a **broader range of system buildouts** than univariate sensitivity analysis (varying one parameter at a time)

Capacity additions
through 2050 [GW]

PV

Wind

Storage

Nuclear

Gas+CCS

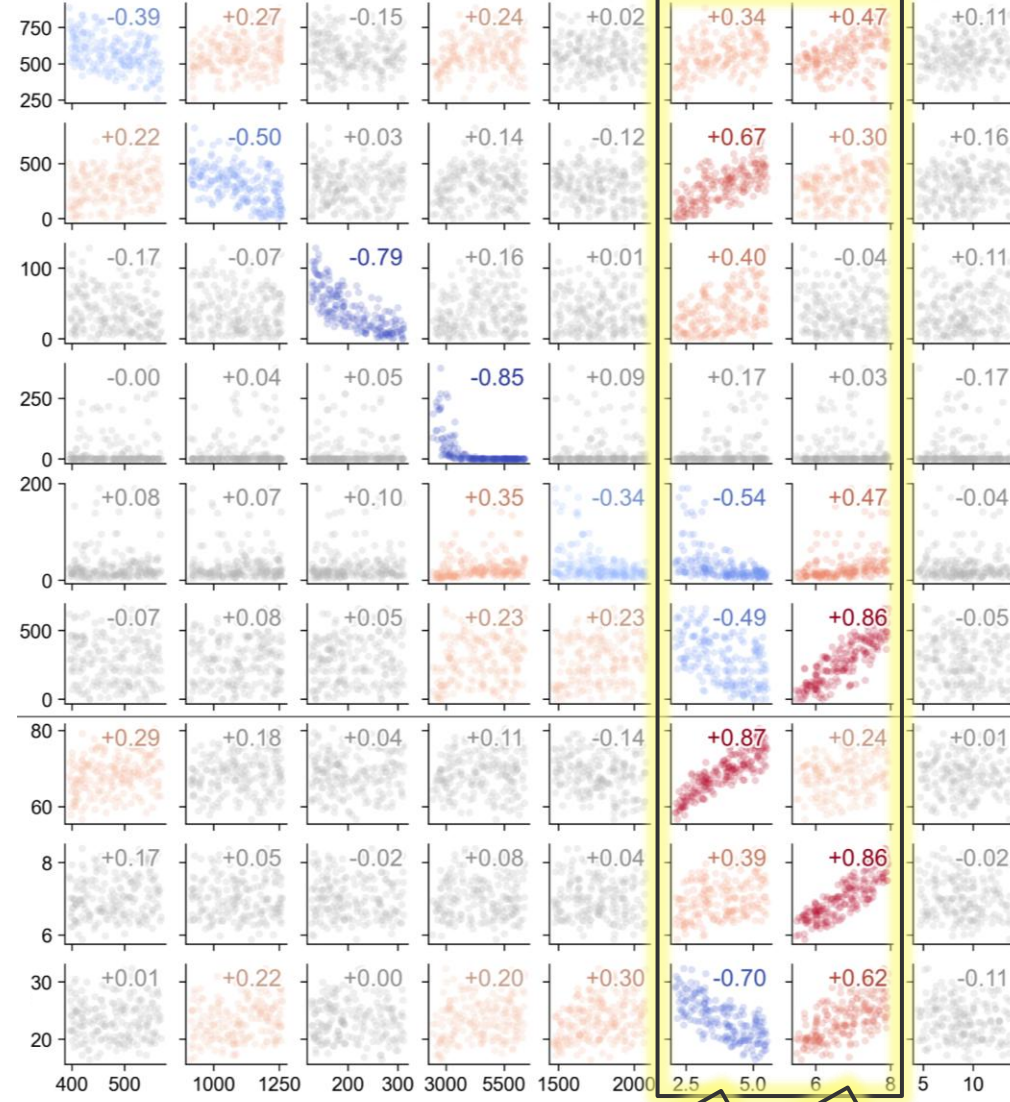
Gas

SCOE
[\$/MWh]

NPC
[\$trillion]

CO₂
[GT]

Spearman rank correlation coefficient:



- Flat distribution
- Country sampling (perfect regional correlation)
- 189 samples

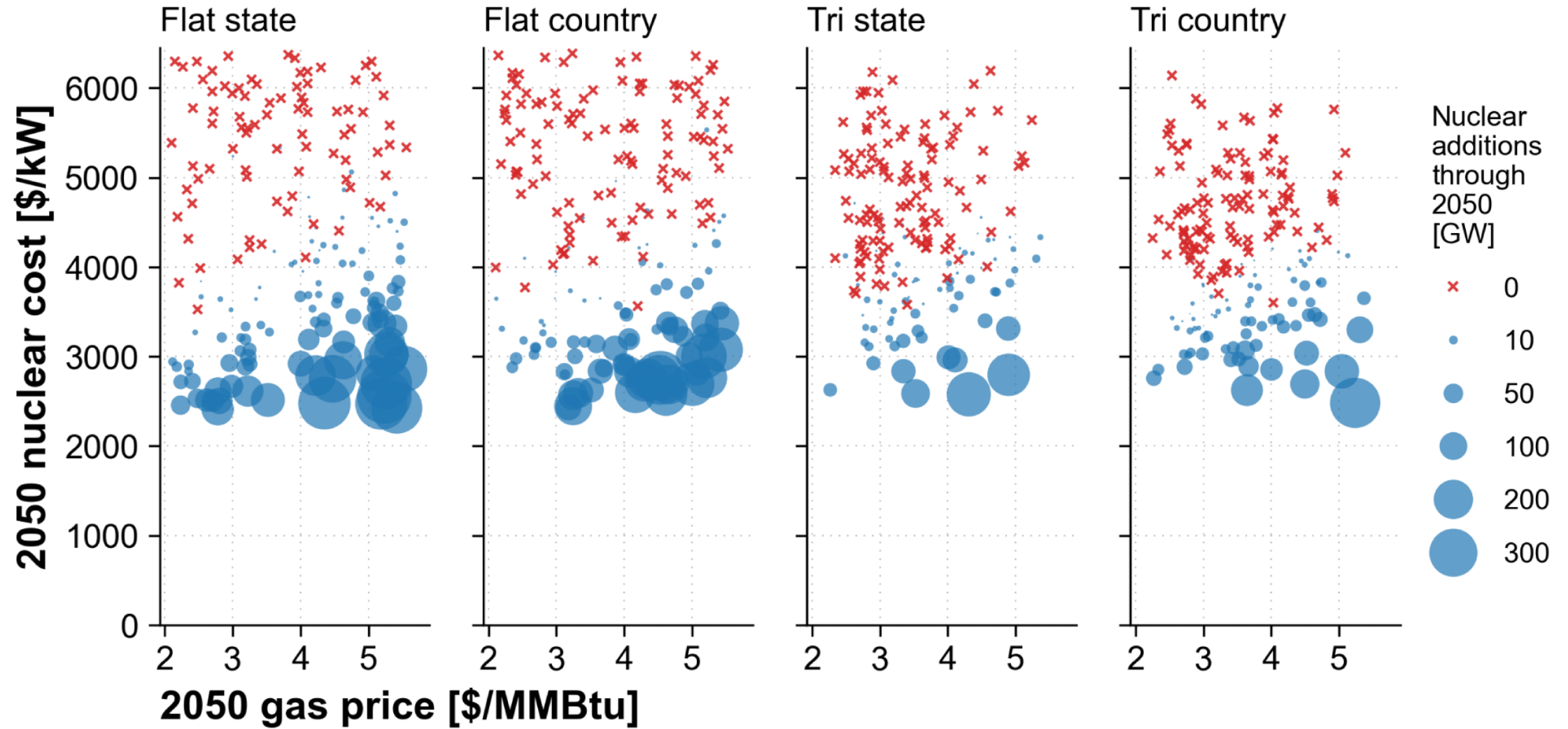
Global sensitivity analysis
(varying all parameters together)
uncovers a **broader range of system buildouts** than univariate sensitivity analysis (varying one parameter at a time)

Helps identify input parameters with the strongest impact on system metrics of interest

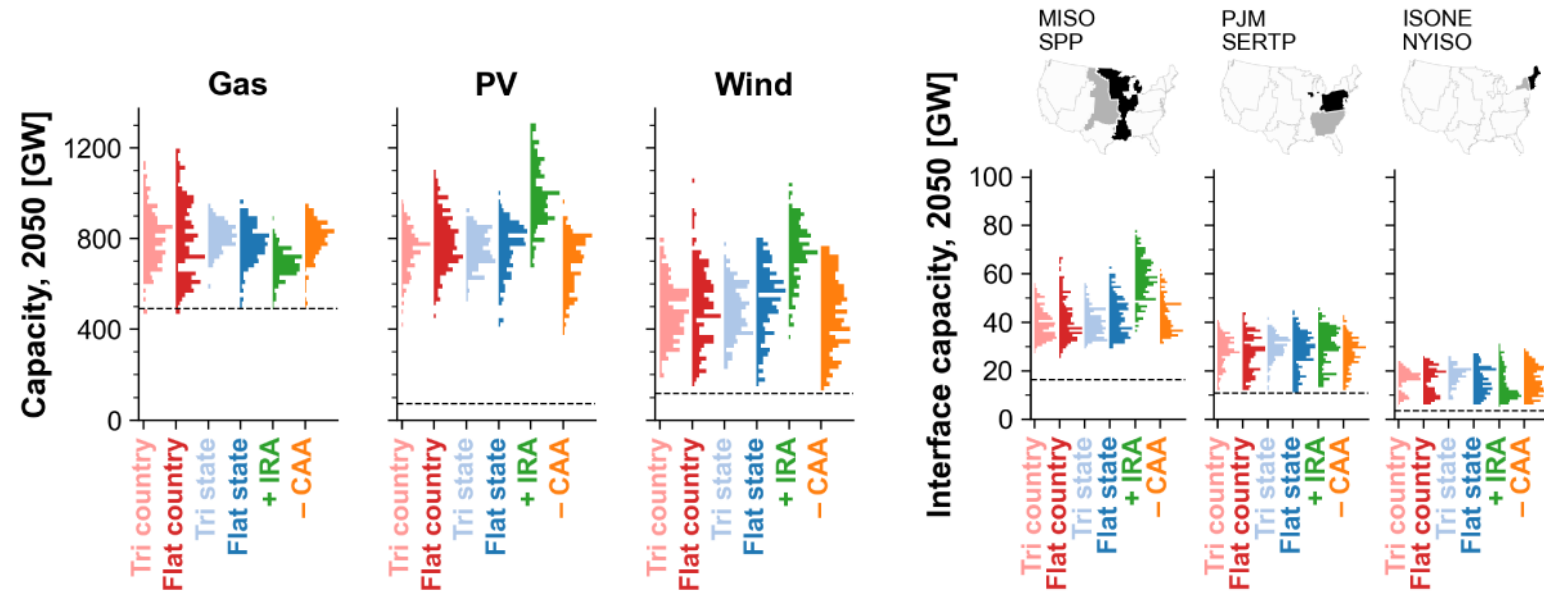
2050:
PV cost [\$/kW]
Wind cost [\$/kW]
Battery cost [\$/kW]
Nuclear cost [\$/kW]
Gas+CCS cost [\$/kW]
Gas price [\$/MMBtu]
Demand [PWh]
Wind avail [TW]

Largest impact on electricity prices, total cost, and emissions

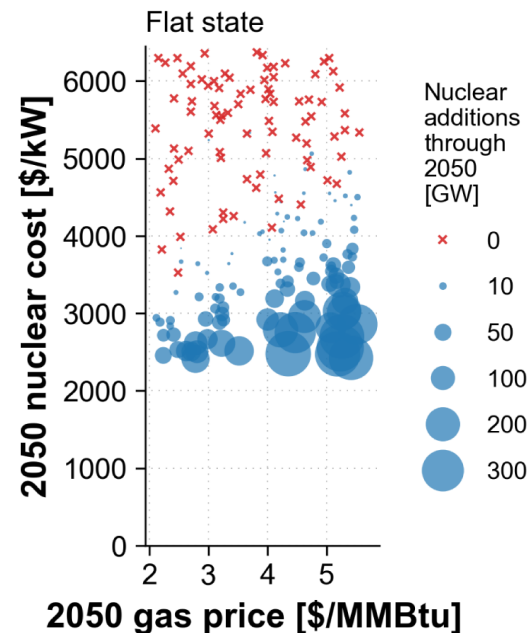
Interactions between inputs



- Typical levels of input parameter uncertainty induce **significant levels of uncertainty** in CEM outputs
- Probability distribution **shape** and regional **correlation** impact the **ranges** of the results, but have **less effect on central values**
 - More regional correlation (e.g., in demand) → more weight in tails



- Results probabilities are determined by input probabilities



- Combining discrete policy scenarios with probabilistic treatment of other inputs provides **uncertainty bounds for policy impact analysis**

